CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Pharm. Examination

University Theory Examination October/November 2016

PH225 Physical Pharmaceutics

Date: 07/11/2016, Monday   Time: 01:30 p.m. to 04:30 p.m.   Maximum Marks: 80

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The interactins responsible for the solubility of ionic crystalline substances in water is
[A] Ion-Dipole
[B] Ion-Induced Dipole
[C] Both A & B
[D] None
2. The temperature at which the vapour pressure of the liquid equals the atmospheric prsssure is known as the
[A] boiling point
[B] melting point
[C] freeziong point
[D] evaporation point
3. Substances exist in more than one crystalline form are said to be
[A] polymorphic
[B] semimorphic
[C] monomorphic
[D] unimorphic
4. The phase rule is expressed as
[A] F=C-P+2
[B] F=P-C+2
[C] F=P-2+C
[D] F=C-2+P
5. Force per unit length existing at the interface between two immiscible liquid phases is called as
[A] interfacial tension
[B] surface tension
[C] facial tension
[D] force tension
6. The full form of HLB is
[A] hydrophilic-lipophilic balance
[B] hydrogen-liason balance
[C] hydrophilic-listed balance
[D] hydro-lio balance

7. The difference in potential between the surface of the tightly bound layer and the electroneutral region of the solution is known as
[A] zeta potential
[B] Nernst potential
[C] both A & B
[D] none

8. The range of particle size for colloidal dispersion is
[A] 1.0 nm to 0.5 μm
[B] 1.0 nm to 5 μm
[C] 10.0 nm to 0.5 μm
[D] 10.0 nm to 5 μm

9. Aggregatres which contains 50 or more monomers of amphiphiles is called as
[A] micelles
[B] solution
[C] mixture
[D] syrup

10. When a strong beam of light passed through a colloidal sol, a visible cone, results due to scattering of light. This effect is called as
[A] Faraday-Tyndall effect
[B] Fday-Tyn effect
[C] Fara-Tydal effect
[D] Fary-Tyna effect

11. The science and technology of small particle is known as
[A] micromeritics
[B] microtics
[C] meritics
[D] metics

12. When the number or weight of particles lying within a certain size ranges is plotted against the size range or mean particle size, a curve obtained is called as
[A] frequency distribution curve
[B] frequently distribution curve
[C] freque distributed curve
[D] fre dis curve

13. Resistance of a fluid to flow is known as
[A] viscosity
[B] density
[C] porosity
[D] viscinity

14. As the temperature increases, viscosity of gas
[A] increases
[B] decreases
[C] remains stable
[D] none of above

15. Systems with high percentage of solids exhibit an increase in resistance to flow with increase in rate of shear is termed as
[A] dilatant
[B] dillan
[C] ditant
[D] ditan

16. To improve flow characteristics, material termed _____ are frequently added to granular powder.
[A] glidants
[B] retardants
[C] tanants
[D] mutants

17. Common types of thermal analysis is _____.
[A] differential scanning calorimetry
[B] differential thermal analysis
[C] thermogravimetric analysis
[D] all of above

18. Solids having definite shape and orderly arrangement of units is called as ________.
[A] Amorphus
[B] Critol
[C] Ampicrysp
[D] Crystal

19. Methods for determining particle size is
[A] optical microscopy
[B] sieving
[C] sedimentation
[D] all of above

20. Derived properties of powder includes
[A] porosity
[B] density
[C] flow properties
[D] all of above

SECTION – II

Q 2	Attempt any **TWO** of the following	
A	Enlist the types of colloids. Expalin any one in detail.	**05**
B	Differentiate between floculated and deflocculated suspension.	**05**
C	Enlist methods of adjustment of isotonicity. Explain any one.	**05**
Q 3	Attempt any **TWO** of the following	
A	Calculate the HLB of Tween 20. Saponification number of the easter, S=45.5 Acid number of the fatty acid, A=276	**05**
B	Find the amount of sodium chloride required to make 500 ml of 1% W/V solution of procaine hydrochloride isotonic with blood plasma. F.P. of 1% W/V solution of procaine hydrochloride is -0.122° C. F.P. of 1% W/V solution of sodium chloride is -0.58° C. F.P. of blood is -0.52° C.	**05**
C	A sample of calcium oxide powder weighing 25 g was found to have a bulk volume of 50 cm^3 and tap volume of 40 cm^3 when place in a graduated cylinder. Calculate bulk density, tap density, Carr's index, housner's ratio and percentage porosity.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Give the working principle of ostwald viscometer with a labelled diagram.	**05**
B	Explain the binding forces exist between molecules.	**05**
C	Expalin inclusion complex in detail.	**05**
D	Write a note on surfactants.	**05**
E	Write a note on zeta potential.	**05**
F	Define suspension. How suspension is evaluated?	**05**

Q 5	Attempt any **FOUR** of the following	
A	Write a note on use of optical microscopy method for measurement of particle size.	**05**
B	Write a note on derived properties of powders.	**05**
C	Write a short note on Gel-Sol-Gel transformation.	**05**
D	Write note on liquid crystals.	**05**
E	Explain the physical stability of emulsion.	**05**
F	Explain following terms:	**05**

1. Stability constant
2. Angle of repose
3. Surface tension
4. Isotonic solution
5. Dilelectric constant
